

BLUE RIDGE, VA, USA

WMO: 745985

Lat: 36.631N

Lon: 80.018W

Elev: 941

StdP: 14.20

Time Zone: -5.00 (NAE)

Period: 95-19

WBAN: 63806

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
1	14.6	18.7	-2.4	5.0	24.0	2.9	6.6	28.2	22.0	34.0	18.9	34.4	2.3	320	0.381

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
7	22.5	93.5	74.3	91.3	74.3	90.0	74.0	78.6	88.4	77.4	87.2	76.2	85.2	3.7	210

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	
75.4	138.0	82.1	74.7	134.8	81.5	73.2	127.6	80.0	42.8	88.5	41.5	86.1	40.3	85.6	85.5

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature									
			Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years			
1%	2.5%	5%	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	DB	WB
(a)	(b)	(c)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(4) 16.8	13.1	10.9	5.5	97.5	6.8	2.7	0.6	99.5	-3.4	101.1	-7.2	102.6	-12.1	104.6		
(5)			4.7	80.8	6.7	2.0	-0.1	82.3	-4.0	83.5	-7.8	84.6	-12.7	86.1		

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
(6) Temperatures, Degree-Days and Degree-Hours	DBAvg	57.8	37.9	41.2	48.1	57.8	65.8	73.5	77.0	75.8	69.1	58.1	47.3	40.5
	DBStd	15.84	9.92	9.09	9.56	8.43	7.13	5.23	4.34	4.73	6.05	7.72	8.60	8.48
	HDD50	1328	394	270	154	24	1	0	0	0	19	150	316	
	HDD65	3931	841	667	528	244	81	7	0	2	31	237	532	761
	CDD50	4160	18	23	95	258	491	705	837	800	573	270	70	20
	CDD65	1287	0	0	4	27	106	262	373	336	154	23	2	0
	CDH74	12895	3	6	90	505	1221	2649	3567	3133	1361	319	35	6
	CDH80	5332	0	0	12	141	411	1133	1662	1415	486	68	4	0
(14) Wind	WSAvg	2.9	3.8	3.9	4.3	4.1	2.8	2.1	1.8	1.7	2.1	2.5	3.0	3.0
(15) Precipitation	PrecAvg	47.9	4.0	2.9	4.5	3.5	4.2	4.8	4.7	4.5	4.3	3.3	3.3	3.7
	PrecMax	66.1	8.5	6.8	8.9	6.2	7.6	8.9	9.2	8.5	12.2	11.2	7.6	8.0
	PrecMin	36.0	0.7	1.1	0.7	0.9	1.3	0.9	2.4	2.0	0.8	0.0	0.6	1.5
	PrecStd	8.1	1.9	1.5	2.0	1.5	2.0	2.2	1.7	1.6	2.8	2.4	1.9	1.4
(19) Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	72.0	73.2	81.6	90.1	91.2	96.8	98.7	97.4	92.7	86.0	78.9	72.9
		MCWB	59.6	60.9	61.3	64.2	71.9	75.1	75.9	75.7	71.8	70.1	62.6	59.3
	2%	DB	64.1	67.5	75.4	83.7	88.3	91.5	93.6	93.2	89.8	81.4	72.5	65.5
		MCWB	56.6	55.0	58.3	63.6	70.6	74.6	75.1	75.2	73.0	66.6	59.5	56.5
	5%	DB	60.6	63.2	71.7	80.6	84.3	90.0	91.2	90.6	85.8	77.4	68.4	61.0
		MCWB	52.4	51.6	56.3	61.7	69.1	74.0	75.4	74.6	71.8	64.3	57.9	53.2
	10%	DB	54.7	57.5	65.9	75.2	81.4	87.6	89.6	88.2	82.0	73.0	63.9	55.8
		MCWB	47.6	47.9	53.6	59.9	67.6	73.0	75.0	74.2	70.4	62.7	55.7	49.2
(27) Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	63.3	62.7	65.1	70.7	75.7	78.5	81.2	80.0	78.1	74.0	67.5	64.2
		MCDB	66.8	68.8	74.7	81.2	85.9	89.2	91.2	88.7	85.8	79.7	73.5	66.9
	2%	WB	59.1	58.8	62.2	67.1	73.3	77.0	79.0	78.3	75.8	71.0	64.2	59.6
		MCDB	63.3	65.4	71.2	78.0	83.6	87.5	89.2	87.9	82.9	75.9	68.5	64.3
	5%	WB	54.2	54.1	59.3	64.7	71.3	75.6	77.5	77.1	74.0	67.8	60.9	54.7
		MCDB	58.6	59.6	67.6	73.6	80.5	85.6	87.5	86.7	81.1	74.3	66.0	58.9
	10%	WB	48.4	49.5	56.4	62.6	69.5	74.2	76.2	75.7	72.2	65.0	56.8	50.2
		MCDB	53.8	57.5	64.1	71.3	77.9	83.6	85.1	84.1	79.3	70.5	62.5	55.0
(35) Mean Daily Temperature Range	5% DB	MDBR	21.8	23.4	25.2	26.9	25.5	24.4	22.5	23.1	22.9	26.2	25.8	22.0
		MCDBR	28.6	31.6	32.5	34.0	30.4	28.6	26.6	27.1	28.9	32.1	32.3	29.4
		MCWBR	19.8	20.3	17.4	15.7	14.4	12.9	10.9	11.5	13.9	17.8	20.0	19.7
	5% WB	MCDBR	25.2	25.4	25.9	26.1	25.3	24.5	22.9	23.4	23.4	24.4	24.5	23.5
		MCWBR	20.1	19.3	15.9	14.0	12.7	11.8	10.3	10.9	12.7	15.6	18.3	18.7
(40) Clear-Sky Solar Irradiance	taub		0.303	0.319	0.346	0.393	0.431	0.474	0.509	0.508	0.405	0.360	0.319	0.306
	taud		2.508	2.453	2.427	2.305	2.217	2.136	2.065	2.043	2.334	2.442	2.516	2.535
	Ebn at Noon		284	292	292	282	272	259	249	246	269	273	275	276
	Edh at Noon		25	30	34	41	45	49	52	52	37	30	25	23
(44) All-Sky Solar Radiation	RadAvg		776	990	1322	1653	1824	1974	1838	1682	1401	1141	862	662
	RadStd		54	120	86	133	152	144	121	87	133	140	70	59

Historical Trends

	DBAvg	Heating		Cooling			Degree-Days			
		99% DB	99% DP	1% DB	1% WB	1% DP	HDD50	HDD65	CDD50	CDD65
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)
(46) Station Only	N/A	N/A	-5.47	N/A	N/A	N/A	N/A	N/A	N/A	N/A
(47) Regional (50 neighbors)	+0.77	N/A	N/A	N/A	N/A	N/A	N/A	N/A	+250	+158

Nomenclature: See separate page